Submission Summary

Conference Name
IEEE Chandigarh Subsection International Conference

Track Name
Artificial Intelligence and Data Science

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Paper Title
Intelligent Automation: Integration of AI-Powered Chatbots and Robotic Process Automation in Enterprise Applications

Abstract
Intelligent automation has become possible for many businesses due to the integration of robotic process automation (RPA) with artificial intelligence (AI). RPA systems are great for performing rule-based tasks, but using RPA systems and AI-enabled chatbots autonomously can put limits on their total capability as a system. To create a holistic automated end-to-end process pipeline, this research proposes a unified framework where RPA and transformer-based conversational AI work together. By processing user input using a multi-stage natural language processing (NLP) approach (i.e., intent identification and entity extraction), the proposed solution can trigger relevant backend workflows automatically.
The framework was tested using a sample of 2,400 annotated customer service transcripts and its accuracy for identifying intent is 94.7%. The total time it took for an agent to respond to customers has decreased from 18.4 seconds to 6.2 seconds and manual intervention rates have dropped from 38.6% to 9.4%. This suggests that AI and RPA are complementary technologies, providing the foundation for scalable, efficient, and intelligent enterprise automation. Additionally, practical deployment issues that influence how these technologies can be successfully utilized include model maintenance, regulatory compliance, and integrating with existing legacy systems.

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Submission Questions Response

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3. Author's Information

I confirm that I have listed the title of the paper and the names, email addresses and affiliations of all contributed authors in the submission system that match the order as given in the submitted paper

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4. Co-author's Consent

I confirm that I have discussed about this submission with all co-authors and authors have approved the paper and agree with its submission to this conference.

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5. Certificate of Originality

I confirm that neither the paper nor any parts of its content are currently under consideration or published in another conference/journal and the paper is free from plagiarism

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6. Author's Consent

It is confirmed that at least one author will register in full to attend and present the paper at the Conference.

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7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

I confirm that no text in the paper is AI-generated.

Agreement accepted